

Meehan and Zhang on Bayes

Cian Dorr

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1. Schoenfield's problem

On a natural interpretation, an “updating rule” is a function f that maps each prior credence function π and possible world w to a probability distribution $f_\pi(w)$ (over the same algebra as π).

- Such a function could capture a disposition an agent could have (the disposition to, at a certain later time, have the credence function that f spits out when fed their current credences and the actualized world), or a plan an agent could make.

Some examples:

- (1) Bayesian conditionalization for agent a at time t : $f_\pi(w) = \pi(\cdot \mid p)$, where p is a 's evidence at t at w .
- (2) Meta-conditionalization for agent a at time t : $f_\pi(w) = \pi(\cdot \mid p)$, where p is $\{w': a$'s evidence at t at w' is the same as a 's evidence at t at $w\}$.
- (3) The omniscient rule: $f_\pi(w)$ is the probability function (on the same algebra as π) such that for all p , $f(w)(p) = 1$ if $w \in p$, otherwise 0.
- (4) The inert rule: $f_\pi(w) = \pi$ for all w .

Fix a *strictly proper scoring rule* A : a function that takes a possible world w and a function π from propositions to $[0,1]$, and returns a number $A(\pi, w)$, in such a way that when π and π' are distinct probability functions, $E_\pi[A(\pi, \cdot)] > E_\pi[A(\pi', \cdot)]$.

Then given any prior probability function π and updating rule f , we can define the **expected accuracy** of f in π as $E_\pi[A(f_\pi(\cdot), \cdot)]$ (i.e. $\sum_w \pi(w) A(f_\pi(w), w)$).

One might hope to devise some notion of what it means for an updating rule to be “followable” by an agent at a time. And one might hope that it would turn out that Bayesian conditionalization (for a , t) has maximal π -expected accuracy within the class of rules followable by a at t .

Whatever followability amounts to, the omniscient rule had better not count as followable (by limited agents for whom it does not coincide with Bayesian conditionalization), since the omniscient rule has higher expected accuracy than any other rule (assuming a regular π , or ignoring measure-zero differences).

A natural proposal: f is followable by a at t iff $f_\pi(w) = f_\pi(w')$ whenever a has the same evidence at t at w and w' .

Schoenfield's problem: according to this criterion, meta-conditionalization is followable. Moreover it has higher expected accuracy than any other followable rule, and hence higher accuracy than Bayesian conditionalization unless it coincides with Bayesian conditionalization. But it does not coincide with Bayesian conditionalization unless evidence is "transparent" (partitional).

2. States and learning situations

The key to Alex and Snow's approach to this puzzle is to split each possible world into two components: a *state* s (belonging to a set of states S with $|S| \geq 2$) and an *learning situation* E .

- The *state* represents the answer to some worldly question we are investigating—it 'captures distinctions that are epistemically relevant to the inquiry at hand'. The agent's evidence is required to be a set of states.
- The *learning situation* is a function that maps each state to a set of states that contains it (let $\mathcal{E}$ be the set of all such functions). *Interpretation*: E is a 's learning situation at t iff for each state s , if s obtained, a 's evidence at t would be $E(s)$.
 - ▶ The setup tends to suggest a view where the two components can be combined freely: for any $s \in S$ and any $E \in \mathcal{E}$ it is possible for the state to be s while the relevant agent's learning situation at the relevant time is E . This is surprising since most members of $\mathcal{E}$ are very strange and random.
- Crucially, *prior and posterior credence functions are defined only over sets of states*. (See section 4 below for some reflections on what could motivate this.)

Reflecting this split, an updating rule will now need to take *three* arguments: a prior π (a probability function on S), an learning situation E , and a state s . They write this as $f_{(\pi, E)}(s)$ (another a probability function on S).

We have analogues of the rules from before:

- (1) BCOND: $f_{(\pi, E)}(s) = \pi(\cdot \mid E(s))$.
- (2) META-COND: $f_{(\pi, E)}(s) = \pi(\cdot \mid E = E(s))$.
- (3) Omniscience: $f_{(\pi, E)}(s)(p) = 1$ if $s \in p$, otherwise 0.
- (4) Inertness $f_{(\pi, E)}(s) = \pi$.

Since priors are only defined on S , a prior doesn't assign an updating rule an expected accuracy scores *simpliciter*. However if we fix an learning situation E as well as a prior π , we can define the **expected accuracy of f in π and E** as $E_\pi[A(\cdot, f_{(\pi, E)}(\cdot))]$.

We define two key properties of update rules:

f is **available** to π given $E :=$ whenever $E(s) = E(s')$, $f_{(\pi, E)}(s) = f_{(\pi, E)}(s')$.

f is **uniformly available** given $\pi :=$ whenever $E(s) = E'(s')$, $f_{(\pi,E)}(s) = f_{(\pi,E')}(s')$.¹

- For any π and E , Omniscience has maximal expected accuracy in π and E among the class of all updating rules. However, Omniscience is not *available* to π given E , unless $E(s)=\{s\}$ for all s .
- META-COND is available to any π given any E , and moreover META-COND has maximal expected accuracy in π and E among the class of updating rules available to π given E . However, META-COND is not *uniformly* available given any regular π .
 - ▶ This is the key to Alex and Snow's response to Schoenfield's challenge. More below on why it might matter.
 - ▶ Note that if we added compatibility conditions, proving that META-COND is not uniformly available would only require compatible E, E', s, s' such that $E(s) = E'(s)$ and $E(s') \neq E'(s')$.
- BCOND and Inertness are uniformly available given any π , as are many other rules.
- For any regular π , there is no rule f uniformly available given π such that for all E , f has maximal expected accuracy in π and E among the class of updating rules uniformly available to π .
- There are pairs of π and E such that BCOND has higher expected accuracy in π and E than Inertness, and pairs for which it's the other way round.

To help understand what is and isn't gained by the restriction to uniformly available updating rules, it is helpful to note that for any fixed learning situation E , we can construct a uniformly available updating rule that agrees with META-COND on E , though not on other learning situations:

E-META-COND: $f_{(\pi,E)}(s) = \pi(\cdot \mid \{s': E(s') = E'(s)\})$ if this set is nonempty, else π .²

- E-META-COND is uniformly available to any π .
- For any π , E-META-COND agrees with META-COND on π and E .
- For any regular π and any E , E-META-COND has maximal accuracy in π and E among the class of updating rules uniformly available to π (and so does any rule that agrees with E-META-COND on π and E). If E is non-transparent, BCOND does not agree with E-META-COND on π and E .

¹ If we thought some combinations of s and E were impossible, it would be natural to restrict these definitions to "compatible" combinations.

² What the rule does in the "else" situation isn't going to be relevant here: an alternative would be to spit back $\pi(\cdot \mid E'(s))$ in that case, as in the rule considered on p. 32.

3. Motivating the restriction to uniformly available updating rules

In earlier drafts of this paper, there was suggestion that non-uniformly-available rules were in some sense not really followable. This suggestion is not present in the current draft. Instead, the main ideas are (a) that the class of uniformly available rules is quite *natural*, and (b) that such a restriction captures a certain ‘anti-holist commitment’.

4. Interpreting the framework

It is crucial to the whole setup that priors and posteriors are defined only on sets of states, rather than on sets of worlds (or pairs of a state and a learning situation). If priors were defined over $S \times \mathcal{E}$, we would have a natural $\mathcal{E}$ -independent notion of “the expected accuracy of an updating rule in π ”, and there would be another presumptive rival to BCOND even among the uniformly available rules, namely the rule that maximizes expected accuracy in this sense.

What is supposed to be the significance of this restriction? I considered three possible interpretations.

Interpretation 1: we are restricting our attention to simple agents who only have the capacity to think about the relevant worldly questions, and cannot have credences in propositions about their own evidence (or in counterfactuals about what their evidence would be in different states).

- ▶ But what would be the interest of such a restriction?
- ▶ Doesn’t the very idea of “planning” to update according to a rule like BCOND require having the concept of evidence?

Interpretation 2: we are restricting our attention to agents who are already certain of the truth about the evidential situation they are in. So their priors over $S \times \mathcal{E}$ can be recovered from their priors over S .

- ▶ This interpretation fits the discussion of the examples, where the agents involved always seem to know their learning situation.
- ▶ One worry is that the restriction seems to leave out a lot of realistic cases. Counterfactuals of the form ‘if the external world were thus and so, my evidence would be E' ’ may be very hard to know (e.g. if there is indeterminism in one’s sensory channels).
- ▶ But there is also a more important worry: it is puzzling why an agent who knew for certain that they are in updating situation E would need to adopt an ‘updating plan’ that prescribes a posterior credence function for pairs s, E' where $E' \neq E$. Shouldn’t such an agent be able to ignore differences between updating plans that only show up in learning situations other than the one they know they are in?

- ▶ If this is right, then even if they are (for whatever reason) limited to only choosing uniformly available updating plans, if they care about accuracy, they are going to choose some rule like E-META-COND that coincides with META-COND on E.

Interpretation 3: the agent may be really be uncertain what evidential situation they are in. But we assume that their credence distribution over evidential situations conditional on any state won't change when they update, so we only need to consider how they will update their credences about the state.

- ▶ First worry: doesn't the assumption that we are limited to updates of this sort (quite apart from the issue of uniform availability) already stack the decks in favor of Bayesian conditioning (which is all about not changing conditional credences), and against meta-conditionalization? After all, if one reacted to getting evidence p by conditioning one's priors over $S \times \mathcal{E}$ on $\{ \langle s, E \rangle : E(s) = p \}$, one's credences conditional on states *would* change.
- ▶ Second worry: if one had prior-credences over evidential situations and got to choose an updating plan with a view to accuracy, shouldn't one be using these priors to combine the E-relative expected accuracies for each updating plan into a single overall expected accuracy score?
 - The results in Section 6 show that given any rich prior (i.e. probability function over $S \times \mathcal{E}$), there is a unique uniformly available updating rule that maximizes expected total accuracy, namely " λ -conditionalization" They also show that only under very special circumstances will this coincide with BCOND.

Based on our discussion over lunch, I feel like none of these interpretations is right—see below for a fourth.

5. Total expected accuracy

Recall:

- For any regular π , there is no rule f uniformly available given π such that for all E, f has maximal expected accuracy in π and E among the class of updating rules uniformly available to π . There are pairs of π and E such that BCOND has higher expected accuracy in π and E than Inertness, and pairs for which it's the other way round.

Snow and Alex suggest that in the light of this, we look at the **total accuracy** of an updating rule f in a prior π , defined as the sum for all $E \in \mathcal{E}$ of the expected accuracy of f relative to π and E.

They show that BCOND has higher total accuracy than any other uniformly available updating rule (given any regular π). And they suggest a 'total accuracy requirement'

according to which this feature makes every rule other than BCOND epistemically irrational.

6. Weighted total accuracy

There is something a bit arbitrary about the concept of total accuracy, and about principles that give it a special role in the theory of rationality. Remember that the members of the set $\mathcal{E}$ is just required to satisfy one minimal requirement on evidence (factivity) — many of them are quite bizarre and don't seem very interesting. Why give equal weight to all of them?

Given a function λ from states and learning situations to real numbers such that for each $s \in S$, $\sum_{E \in \mathcal{E}} \lambda(s, E) = 1$, one can define a more general notion of the **weighted total accuracy** of f in π given λ : $\sum_{s \in S} \sum_{E \in \mathcal{E}} \pi(s) \lambda(s, E) A(f_{(\pi, E)}(s), s)$.

- One possible interpretation of λ is that we really do have “rich” priors defined on $S \times \mathcal{E}$: $\lambda(s, E)$ is just $\pi(\{E\} \mid \{s\})$.
- A very different interpretation is that it is a primitive normative input measuring the “importance” of updating accurately when it comes to different propositions.

For any given λ , there is a uniformly available updating rule that has higher λ -weighted total accuracy than any other (given any regular π), namely “ λ -conditionalization”: a rule on which one responds to evidence p by “conditionalizing” one's credences over S by the random variable p_λ whose value on any s is $\sum_{E: E(s)=p} \lambda\{s, E\}$.

Only for very special (‘balanced’) λ s will λ -conditionalization coincide with BCOND.

- If we were interpreting λ as a primitive normative input, we could take this to be a surprising discovery: insofar as you think Bayesian conditionalization is rationally required and also accept some sort of accuracy requirement, you had better accept that the primitive normative inputs work like that.
- If we were interpreting λ in terms of a rich prior, balancedness looks very hard to motivate.

7. More on the interpretation of the framework

Our conversation over lunch suggested a different vision: real agents often know what learning situation they are in, but we understand the “rational” updating rule as the one that would be chosen “behind the veil of ignorance” by agents ignorant of their learning situation. The weights λ describe that hypothetical state of ignorance.

- One worry about this: it seems to concede that agents who do know their learning situation should prefer, for accuracy's sake, to follow some *irrational* rule. What's the point of the notion of rationality if rational agents should do their best to stop being rational in order to be closer to the truth?